LN8820 Datasheet

A Single-Chip Wireless MCU for Wi-Fi and IOT Applications

Revision 1.9

Jul 9th, 2020

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Contents

1 Overview

1.1 Functional Features

- Wifi Features
  - 2.4GHz 802.11 b/g/n
  - Support STA/AP/STA+AP operation modes
  - Support WPA/WPA2
  - 18dBm output power in 11b mode
  - -94dBm sensitivity
- High Integration
  - Integrated ARM© Cortex - M4F
  - Integrated TCP/IP protocol stack
  - Integrated TR switch, balun, LNA, power amplifier and matching network
  - High precision oscillator
  - AES-128\AES-192\AES-256 hardware encryption
  - 256 bits EFUSE
- Interface
  - Single supply voltage with wide range supported (2.6~3.6V)
  - SPI/SDIO/UART interface
  - SPI/I2C master for sensor
  - I2S, PWM and GPIO
  - Multi-channel ADC
  - XIP from external QSPI FLASH with high-speed cache
- Low Power
  - 2.5mA in standby mode (DTIM=1, keep connected with AP)
  - 50uA in deep sleep mode(fast wakeup)
  - 20uA in retention mode
- MCU
  - 40/80/160M clock for M4F
  - 296KB on-chip SRAM
  - Up to 4MB Flash for XIP application code
  - SWD debug interface
- Package
  - QFN40 5mm*5mm

1.2 Block diagram

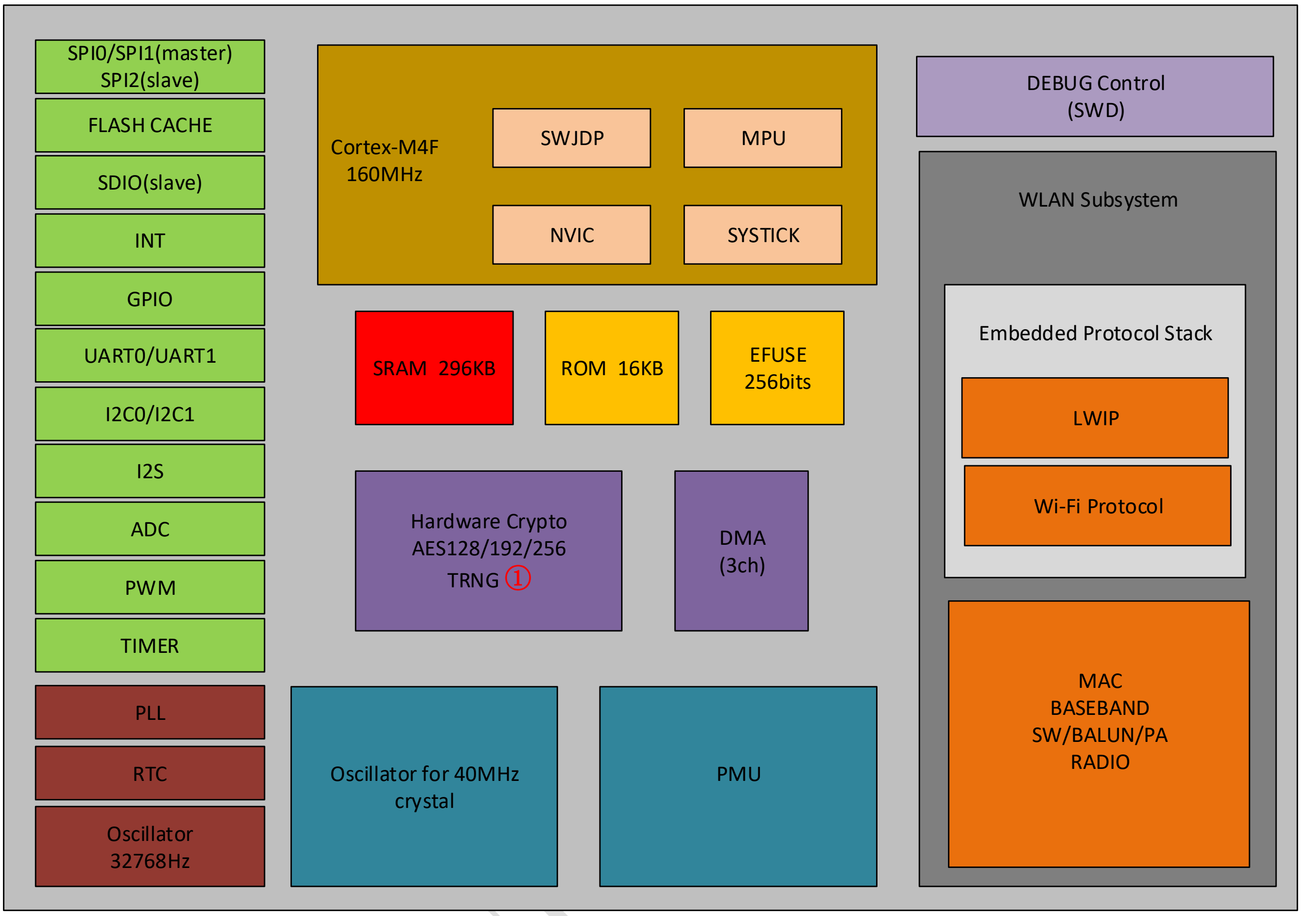


Figure 1-1 LN8820 block diagram

Notes:

① TRNG stands for True Random Number Generator

1.3Memory Map

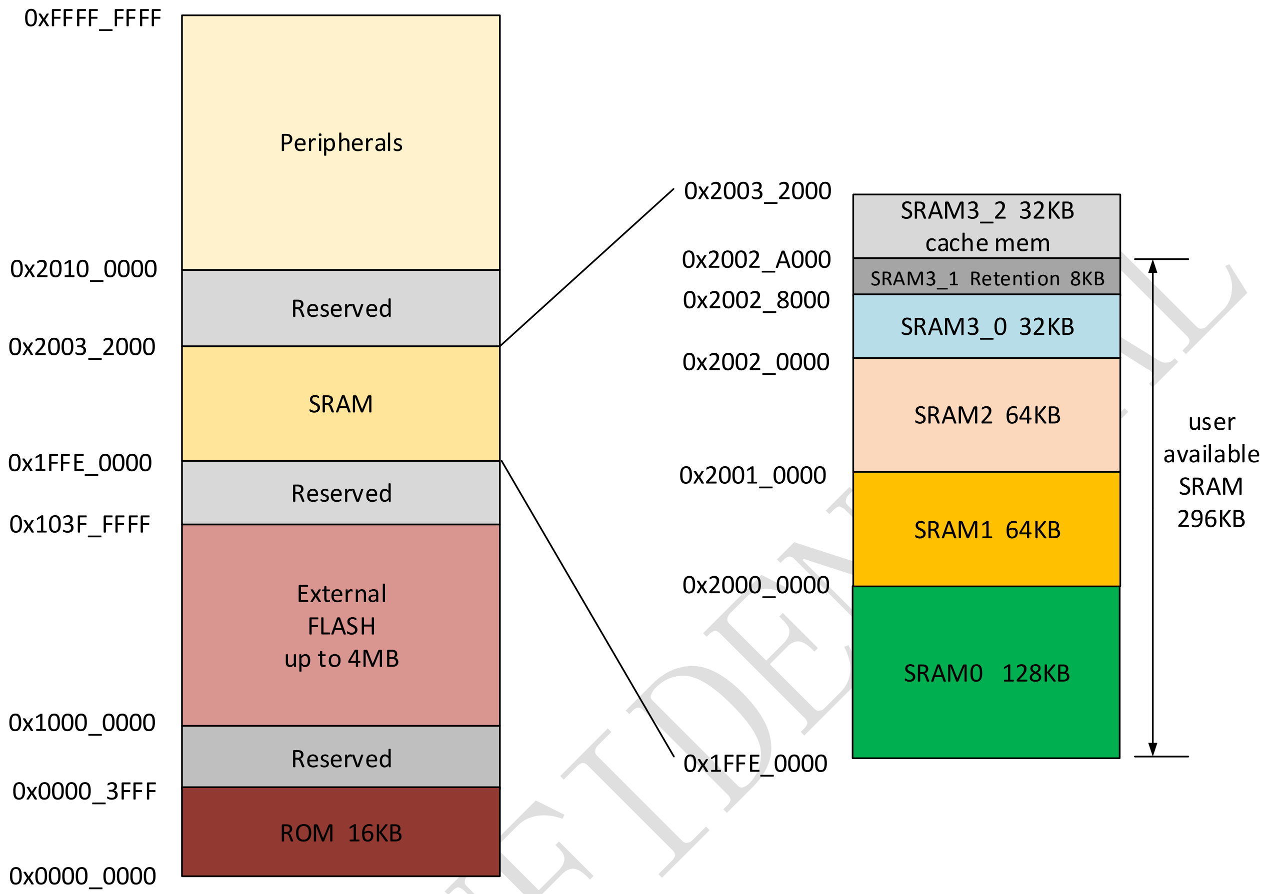


Figure 1-2 LN8820 Memory Map

2 Wi-Fi Subsystem

2.1 Supported Frequencies

Table 2-1 Supported Frequencies

Parameter	Condition	Min	Typ	Max	Unit
Receive frequency range 2.4GHz		2412		2484	MHz

2.2 Receiver Characteristics

Table 2-2 Receiver Characteristics

Parameter	Condition	Min.	Typ.	Max.	Unit
Sensitivity					
11b, 1M	FER<8%, 1024 bytes		-94		dBm
11b, 11M	FER<8%, 1024 bytes		-88		dBm
11g, 6M	FER<10%, 1024bytes		-90		dBm
11g, 54M	FER<10%, 1024 bytes		-74		dBm
11n, MCS0	FER<10%, 1024 bytes		-90		dBm
11n, MCS7	FER<10%, 1024 bytes		-71		dBm
Maximum input level					
11b	FER<8%, 1024 bytes		4		dBm
11g	FER<10%, 1024 bytes		-6		dBm
11n	FER<10%, 1024 bytes		-6		dBm
Operating power consumption					
11b			63		mA
11g			68		mA
11n			68		mA

2.3 Transmitter Characteristics

Table 2-3 Transmitter Characteristics

Parameter	**Condition**	**Min.**	**Typ.**	**Max.**	**Unit**
Output power					
11b, 1M DSSS	Maximum RMS output power measured		18	19	dBm
11g, 54M OFDM	Maximum RMS output power measured		14	15	dBm
11n, MCS7	Maximum RMS output power measured		13	14	dBm
Power consumption					
11b	Continuous transmitting		260		mA
11g	Continuous transmitting		220		mA
11n	Continuous transmitting		220		mA

3 Power Management

3.1 Power Mode

There are 5 power modes to manage lower power operation of the system.

- **Active:** Normal mode
  SW can gate clock of some Peripherals.
- **LightSleep:** CPU executes WFI instruction without 'SLEEPDEEP',and gates clock by itself
  Other modules are still in free running.
  CPU exit this mode from wakeup by interrupt.
- **DeepSleep:** CPU executes WFI instruction with 'SLEEPDEEP'.
  All digital parts are all clock gated except AWO[1] domain but CPU power is still on.
  Voltage supplied to Digital parts can be reduced to save power.
  Clock in AWO is switched to 32KHz
  WIC wakes up CPU, then all clocks and power recover to normal state.
- **Retention:** CPU executes WFI instruction with 'SLEEPDEEP'
  All digital parts are power-down except AWO domain and selected retention SRAM.
  Registers inside CPU can be retained
  Clock in AWO will be switched to 32KHz
  WIC wakeup CPU, then all clock and power recover to normal state.

Any low power mode can't be switched to another low power mode directly,it must be switched to active mode before going to another one.

Clock and power gating of one domain should be configured before executing WFI.

Power Save Mode Current

Table 3-1 Current in WiFi Power Save Mode

Mode	Condition	Min	Typ	Max	Unit
Power Save Mode	DTIM=1[2]		2.5		mA
	DTIM=3		1		mA

Notes:

[1]AWO: The block whose power is never down is named as Always On

[2]Test condition:keep connected with AP

DTIM=n means STA wakes up to listen to beacon from AP in every n times of DTIM interval

3.2 Power Supplies

Table 3-2 Power Supplies

Power Domain	Description	Min	Typ.	Max	Unit
VDD33	Supply for all chips except PA	2.6	3.3	3.6	V
VDDA33	Supply for PA	2.6	3.3	3.6	V
VIO1	Supply for I/O	2.6	3.3	3.6	V
VIO2	Supply for I/O	2.6	3.3	3.6	V

4 Package Specification

4.1 Pin Laylout

LN8820 is assembled 5mm x 5mm QFN package of 40-pin with 0.4mm pitch.

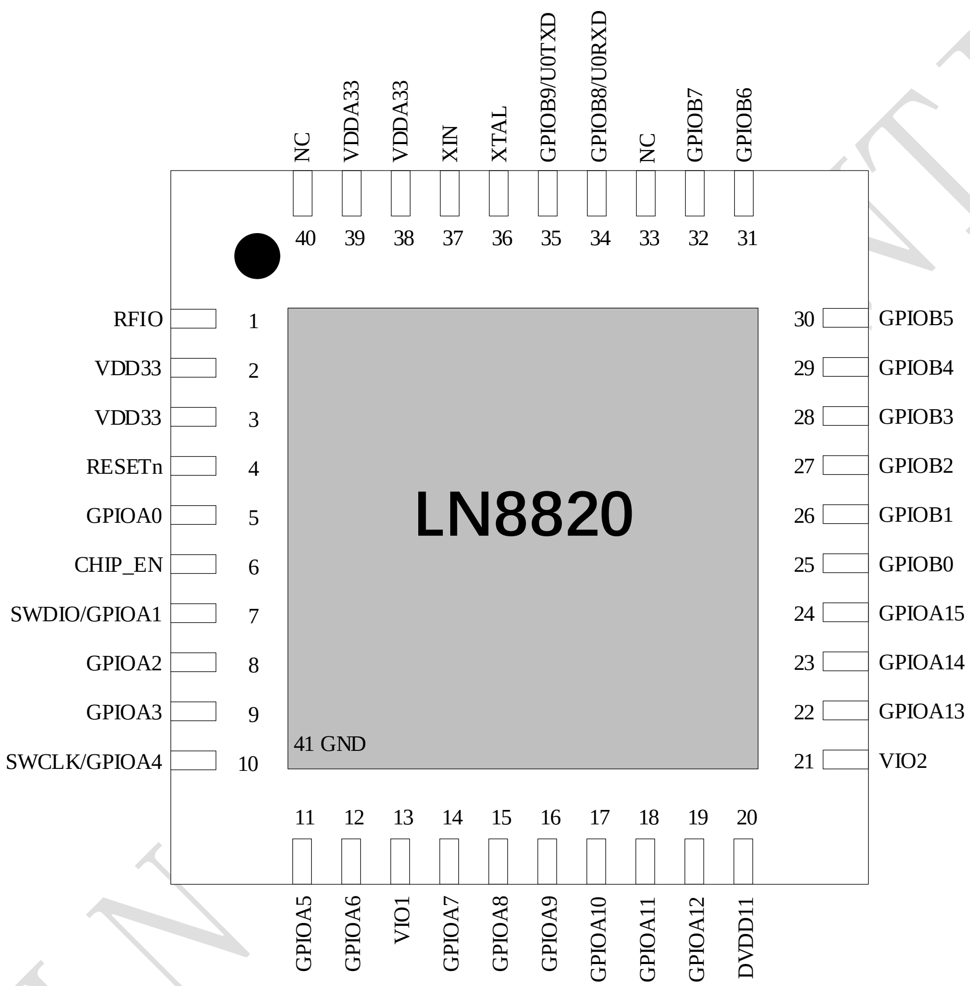


Figure 4-1 LN8820 Pin Layout

4.2 Pin Description

LN8820 totally has 26 General Purpose Inputand Outputs (I/O).

- Two group IOs GPIOA and GPIOB,which are separately configurable
- Configurable hardware and software control for each signal
- Configurable interrupt mode for Port A,debounce logic for interrupts,Generate single hi-active interrupts

4.2.1 Programmable I/Os

Table 4-1 Programmable I/Os

Pin Num	Pin Name	Normal Mode (after reset)	Logic Level	PU / PD / HiZ 110K/after reset	IN/OUT Put (after reset)
5	GPIOA0	GPIOA0	VIO1	PU	IN
7	GPIOA1	SWDIO①	VIO1	PU	IN
8	GPIOA2	GPIOA2	VIO1	PU	IN
9	GPIOA3	GPIOA3	VIO1	PU	IN
10	GPIOA4	SWCLK②	VIO1	DOWN	IN
11	GPIOA5	GPIOA5	VIO1	PU	IN
12	GPIOA6	GPIOA6	VIO1	DOWN	IN
14	GPIOA7	GPIOA7	VIO1	DOWN	IN
15	GPIOA8	GPIOA8	VIO1	DOWN	IN
16	GPIOA9	GPIOA9	VIO1	PU	IN
17	GPIOA10	GPIOA10	VIO1	PU	IN
18	GPIOA11	GPIOA11	VIO1	DOWN	IN
19	GPIOA12	GPIOA12	VIO1	PU	IN
22	GPIOA13	GPIOA13	VIO2	PU	IN
23	GPIOA14	GPIOA14	VIO2	PU	IN
24	GPIOA15	GPIOA15	VIO2	PU	IN
25	GPIOB0	GPIOB0	VIO2	PU	IN
26	GPIOB1	GPIOB1	VIO2	PU	IN
27	GPIOB2	GPIOB2	VIO2	PU	IN
28	GPIOB3	GPIOB3	VIO2	DOWN	IN
29	GPIOB4	GPIOB4	VIO2	PU	IN
30	GPIOB5	GPIOB5	VIO2	PU	IN
31	GPIOB6	GPIOB6	VIO2	PU	IN
32	GPIOB7	GPIOB7	VIO2	PU	IN
34	GPIOB8	UART0_RX③	VIO2	PU	IN
35	GPIOB9	UART0_TX④	VIO2	PU	IN

Notes:

(1) ①② is gpio mode after system reset and can be configured as SWD by bootrom program

(2) ③④ is gpio mode after system reset and can be configured as UART0 by flash program.

4.2.2 Power,Clocks and Reset I/Os

Table 4-2 Power,Clocks and Reset I/Os

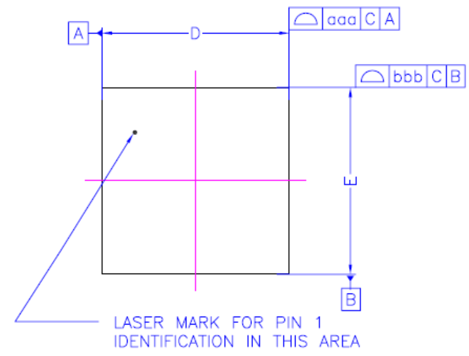
Pin Num	Pin Name	Pin description
1	RFIO	Wi-Fi RF transmitter/receiver
2	VDD33	3.3v power supply
3	VDD33	3.3v power supply
4	RESETn	system reset
6	CHIP_EN	chip enable
13	VIO1①	I/O power supply 1
20	DVDD11	1.1v power output
21	VIO2②	I/O power supply 2
37	XIN	Crystal input
36	XTAL	Crystal output
38	VDDA33	power supply for PA
39	VDDA33	power supply for PA
33	**NC**	not connect
40	**NC**	not connect

Notes:

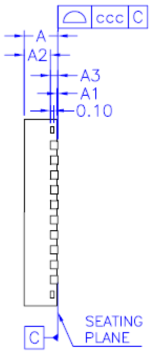
(1) ① ②VIO1,VIO2 must be supplied externally

(2) ②VIO2 should be the same as the voltage supplied for SPI flash, 3.3V is recommended

4.3 Package Information



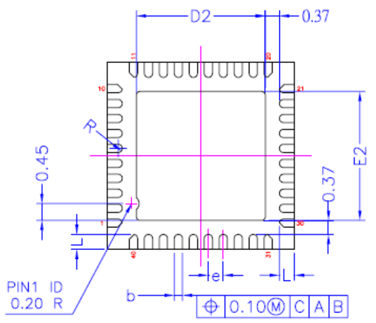
TOP VIEW



SIDE VIEW

* CONTROLLING DIMENSION : MM

SYMBOL	MILLIMETER			INCH		
	MIN.	NOM.	MAX.	MIN.	NOM.	MAX.
A	0.80	0.90	1.00	0.031	0.035	0.039
A1	---	---	0.05	---	---	0.002
A2	---	0.70	---	---	0.028	---
A3	0.20 REF.			0.008 REF.		
b	0.15	0.22	0.27	0.006	0.009	0.011
D	5.00 bsc			0.197 bsc		
D2	3.35	3.45	3.55	0.132	0.136	0.140
E	5.00 bsc			0.197 bsc		
E2	3.35	3.45	3.55	0.132	0.136	0.140
L	0.35	0.40	0.45	0.014	0.016	0.028
e	0.40 bsc			0.016 bsc		
R	0.08	---	---	0.003	---	---
TOLERANCES OF FORM AND POSITION						
aaa	0.10			0.004		
bbb	0.10			0.004		
ccc	0.05			0.002		



BOTTOM VIEW

NOTES :
1.ALL DIMENSIONS ARE IN MILLIMETERS.
2.DIE THICKNESS ALLOWABLE IS 0.305 mm MAXIMUM(.012 INCHES MAXIMUM)
3.DIMENSIONING & TOLERANCES CONFORM TO ASME Y14.5M. −1994.
4.THE PIN #1 IDENTIFIER MUST BE PLACED ON THE TOP SURFACE OF THE PACKAGE BY USING INDENTATION MARK OR OTHER FEATURE OF PACKAGE BODY.
5.EXACT SHAPE AND SIZE OF THIS FEATURE IS OPTIONAL.
6.PACKAGE WARPAGE MAX 0.08 mm.
7.APPLIED FOR EXPOSED PAD AND TERMINALS. EXCLUDE EMBEDDING PART OF EXPOSED PAD FROM MEASURING.
8.APPLIED ONLY TO TERMINALS.

TITLE	PACKAGE OUTLINE 40L DQFN QFN 5.0x5.0 mm	
UNIT	TOLERANCE	
	DIMENSION	ANGLE
INCH / MM		

5 Peripheral Interface

5.1 SWD

LN8820 supports Serial Wire Debug (SWD) which is used to download flash image and debug.The pins shown in Table 5-1 are assigned to SWD mode by bootrom program after power on.

Table 5-1 SWD Pin Assignment

Pin Num	Pin Name	Function Name
7	GPIOA1	SWD
10	GPIOA4	SWCK

5.2 ADC

LN8820 has a 6-channel ,12-bit sampling Analog-to-Digital Converter (ADC). The main features:

- Up to 6 external channels
- Supply voltage and internal temperature is also measured by the ADC
- Support up to 8 result FIFO with selectable FIFO depth
- Conversion complete flag and interrupt
- Selectable asynchronous hardware conversion trigger
- Automatic comparison against programmable value to generate interrupt

Table 5-2 ADC Pin Assignment

Pin Num	Pin Name	Function Name
5	GPIOA0	ADC CH0
7	GPIOA1	ADC CH1
10	GPIOA4	ADC CH2
28	GPIOB3	ADC CH3
29	GPIOB4	ADC CH4
30	GPIOB5	ADC CH5

5.3 SDIO

LN8820 features one slave sdio interface which supports the standard SDIO version 2.0 and provides high-speed data I/O with low power consumption for mobile electronic devices.The main features:

- DMA supported
- The clock is up to 40MHz
- Supported 1-bit,4-bit data mode

Table 5-3　SDIO Pin Assignment

Pin Num	Pin Name	Function Name
12	GPIOA6	SDIO_IO2
14	GPIOA7	SDIO_IO3
15	GPIOA8	SDIO_CMD
16	GPIOA9	SDIO_CLK
17	GPIOA10	SDIO_IO0
18	GPIOA11	SDIO_IO1

5.4 QSPI

LN8820 supports one QSPI interface which is generally used to communicate with flash.The clock of QSPI is programmable and maximum at 80 MHz.

Table 5-4　QSPI Pins Assignment

Pin Num	Pin Name	Funtion Name
22	GPIOA13	QSPI_IO3
23	GPIOA14	QSPI_IO2
24	GPIOA15	QSPI_CS
25	GPIOB0	QSPI_CLK
26	GPIOB1	QSPI_IO1
27	GPIOB2	QSPI_IO0

5.5 I2S

LN8820 features one slave I2S interface,which is generally used to handle the transfer of audio data. The pin definition of I2S is shown in Table 5-5.The main features are described as below.

- I2S transmitter and/or receiver based on the Philips I2S serial protocol
- Slave mode of operation
- 1 channels for both transmitter and receiver
- APB data bus widths of 8, 16, and 32 bits
- Audio data resolutions of 12, 16, 20, 24, and 32 bits
- External sclk gating and enable signals
- Support for programmable DMA registers

Table 5-5 I2S Pin Assignment

Pin Num	Pin Name	Function Name
12	GPIOA6	I2S_SDI0
15	GPIOA8	I2S_WS
16	GPIOA9	I2S_CLK
17	GPIOA10	I2S_SDO0

5.6 Interrupt

There are 16 interrupt sources from external pins and 8 interrupt sources that can wake up the system from sleep.These interrupts support both edge and level mode.The pins assignment of interrupt are listed in Table 5-6.

Table 5-6 Interrupt Pin Assignment

Pin No.	Pin Name	Function 9(EXT INT)	Wakeup INT
5	GPIOA0	EXT INT0	WAKEUP INT[0]
7	GPIOA1	EXT INT1	WAKEUP INT[1]
8	GPIOA2	EXT INT2	WAKEUP INT[2]
9	GPIOA3	EXT INT3	WAKEUP INT[3]
10	GPIOA4	EXT INT4	
11	GPIOA5	EXT INT5	WAKEUP INT[4]
12	GPIOA6	EXT INT6	WAKEUP INT[5]
14	GPIOA7	EXT INT7	WAKEUP INT[6]
15	GPIOA8	EXT INT8	
16	GPIOA9	EXT INT9	
17	GPIOA10	EXT INT10	
18	GPIOA11	EXT INT11	
19	GPIOA12	EXT INT12	
22	GPIOA13	EXT INT13	

23	GPIOA14	EXT INT14	
24	GPIOA15	EXT INT15	
35	GPIOB9		WAKEUP INT[7]

5.7 Boot Mode

LN8820 has two boot modes that depend on the logic value set of GPIOA8,GPIOA9 and GPIOA10 during power-on stage.GPIOA8 is pulled down while GPIOA9 and GPIOA10 are pulled up internally.The pin assignment is listed in Table 5-7 and boot mode configuration is shown in Table 5-8.

Table 5-7 Boot Mode Pin Assignment

Pin No.	Pin Name	Function 10(Boot Mode)
15	GPIOA8	BOOT_MODE[0]
16	GPIOA9	BOOT_MODE[1]
17	GPIOA10	BOOT_MODE[2]

Table 5-8 Definition of Boot Mode

MODE	GPIOA8	GPIOA9	GPIOA10	Description
FLASH Boot	X	X	1	DEFAULT: GPIO8 pull-down GPIO9 and GPIO10 pull-up internally.
UART Boot	0	1	0	CPU will wait data from UART to program flash.

5.8 Alternate Function I/O

The pins are named as alternate function I/O ,which can be configured to any function mode at one time. There are 20 alternate functional I/Os referred to Table 5-9 and 38 function modes shown in Table 5-10.

Table 5-9 Alternate function I/Os

Pin No.	Pin Name	Function 7(Alternate Function IO)
5	GPIOA0	AltFunIO[0]
7	GPIOA1	AltFunIO[1]
8	GPIOA2	AltFunIO[2]
9	GPIOA3	AltFunIO[3]
10	GPIOA4	AltFunIO[4]
11	GPIOA5	AltFunIO[5]
12	GPIOA6	AltFunIO[6]
14	GPIOA7	AltFunIO[7]

15	GPIOA8	AltFunIO[8]
16	GPIOA9	AltFunIO[9]
17	GPIOA10	AltFunIO[10]
18	GPIOA11	AltFunIO[11]
19	GPIOA12	AltFunIO[12]
28	GPIOB3	AltFunIO[13]
29	GPIOB4	AltFunIO[14]
30	GPIOB5	AltFunIO[15]
31	GPIOB6	AltFunIO[16]
32	GPIOB7	AltFunIO[17]
34	GPIOB8	AltFunIO[18]
35	GPIOB9	AltFunIO[19]

Table 5-10 Alternate functions

Function Num	Function Mode	Function Num	Function Mode
0	I2C0_SCL	19	PWM5
1	I2C0_SDA	20	PWM6
2	I2C1_SCL	21	PWM7
3	I2C1_SDA	22	PWM8
4	UART0_TX	23	PWM9
5	UART0_RX	24	PWM10
6	UART0_RTS	25	PWM11
7	UART0_CTS	26	SPI1_SCK⑤
8	UART1_TX	27	SPI1_CS
9	UART1_RX	28	SPI1_MOSI
10	TIMER_PWM0	29	SPI1_MISO
11	TIMER_PWM1	30	SPI2_CLK⑥
12	TIMER_PWM2	31	SPI2_CS
13	TIMER_PWM3	32	SPI2_MOSI
14	PWM0	33	SPI2_MISO
15	PWM1	34	SPI0_CLK⑦
16	PWM2	35	SPI0_CS
17	PWM3	36	SPI0_MOSI
18	PWM4	37	SPI0_MISO

Note:

(1). ⑤⑦SPI0 and SPI1 only support master mode;

(2). ⑥SPI2 only supports slave mode

5.9 I2C

LN8820 has two I2C interfaces which both support master and slave mode. Any two pins listed in Table 5-9 can be configured as I2C.Their features:

• Standard, Fast and high-speed mode are supported

• Master or slave I2C operation

• 7 or 10-bit addressing, 7 or 10-bit combined format transfers

• Bulk transmit mode, DMA request signals

• Programmable slave address,default slave address :0x055

• Interrupt or polled-mode operation

• Programmable SDA hold time

5.10 UART

There are 2 UARTs in LN8820,UART0 and UART1.UART0 supports hardware flow control,which is generally used to output log .While UART1 is suggested to used for transferring data,such as AT command. Any pin listed in Table 5-9 can be configured as UART.Their common features:

• 16 bytes Transmit and receive FIFOs.

• DMA supported

• Shadow registers to reduce software overhead

• Transmitter Holding Register Empty (THRE) interrupt mode

• Functionality based on the 16550 industry standard:

• Programmable character properties, such as number of data bits per character (5-8), optional

• parity bit (with odd or even select) and number ofstop bits (1, 1.5 or 2)

• Line break generation and detection

• Prioritized interrupt identification

• Programmable serial data baud rate (up to 2000000 B/S)

5.11 SPI

There are three independent SPI interface: SPI0 and SPI1 are masters, and SPI2 is slave. Any pin listed in Table 5-9 can be configured as dedicated function for SPI. There are no FIFO or no DMA supported in SPI1.

features:

• 4-wire serial interface: SCK CSn SDI SDO

• 16 words Tx and Rx fifo(SPI0/SPI2)

• DMA interface supported(SPI0/SPI2)

• Dynamic control of the serial bit rate of the data transfer.The maximum frequency of the bit-rate clock (sclk_out) is half of APBCLK

5.12 PWM

There are 4 PWMs inside Timer which has 24-bit width counter .Another 12 PWMs with 16-bit width counter are integrated.The clock source is from APB clock. Duration of low level and high level need to be set, and it determines the duty cycle of PWM waveform. Pins listed in Table 5-9 can be configured as output of PWM.

6 Electrical Characteristics

6.1 Absolute Maximum Ratings

Table 6-1 Absolute Maximum Ratings

Symbol	Min.	Max.	Unit
Supply Voltage	0	3.6	V
Logical Voltage	0.3*VIO	VDD33+0.3	V
Storage Temperature	-40	120	ºC

6.2 Recommended Operating Conditions

Table 6-2 Recommended Operating Conditions

Symbol	Symbol	Min.	Typ.	Max.	Unit
Power Supply	VDD33	2.6	3.3	3.6	V
VIO	VIO1/VIO2	2.6	3.3	3.6	V
Operating Temperature		-40	25	110	ºC

6.3 Input/Output Terminal Characteristics

Table 6-3 Input/Output Terminal Characteristics

Characteristic	Min.	Typ.	Max.	Unit
Input low voltage V_{IL}	-0.3	0	0.6	V
Input high voltage V_{IH}	VIO-0.6	VIO	VIO+0.3	V
Output low voltage V_{OL}	-0.45	0	0.45	V
Output high voltage V_{OH}	VIO-0.5	VIO	VIO+0.5	V

7 Application Notes

7.1 Schematic Guide

1) GPIOA8,GPIOA9 and GPIOA10 need to be kept unconnected or pulled according to different boot mode
2) CHIP_EN must be pulled or driven to high to enable the chip. It can be used to power down the entire chip by pulling it low
3) The VIO must be supplied externally or tied with the VDD33
4) Confirm the key components,refers to Table 7-1
5) RESETn is recommended to be connected to ground through a 2.2nF capacitor

Table 7-1 Key Components

Item	Part Number	Description	Vendor	Comment
Crystal	TZ0308D	SMD;40M;3.2x2.5mm; -30C ~ +85C;CL=9Pf;+/-10ppm	TST	
Flash	GD25Q16ETIG	16Mb;3.0V;	GD	High-temperature
Flash	XT25F16B	16Mb;3.0V;	XTX	Normal-temperature

7.2 Layout Guide

1) 4 Layers is recommended
2) Set Ground Plane closed to component plane
3) Keepout the crystal signal PINs image on Ground plane
4) No slot on the RF return path
5) Keep the power trace width above 15mil
6) Keep the complex impedance of RF path 50 Ohm

8 Revision History

Table 8-1 Revision History

Version	Changlist	Name	Date
0.1	Draft	Jack	2019/08/28
1.0	Original Version	Shaohe.huan	2019/09/10
1.1	LN8820 first version separated from LN882X	Shaohe.huang	2019/10/15
1.2	Pin 33~37 layout changes	Shaohe.huang	2019/10/28
1.3	Power mode ,sdio and application notes description changes	Shaohe.huang	2019/10/29
1.4	Rearrange chapter,update parameters of wifi and power	Shaohe.huang	2019/12/06
1.5	1.Update wifi receive characteristics parameter; 2.Chapter 1.3 Memory Map error correction	Shaohe.huang	2019/12/25
1.6	Update wifi transmitter characteristics parameter;	Shaohe.huang	2020/01/07
1.7	Update wifi transmitter and receiver characteristics parameter	Shaohe.huang	2020/03/10
1.8	Update wifi receiver characteristics parameter	Shaohe.huang	2020/03/27
1.9	Pin layout modify	Shaohe.huang	2020/07/09

